



Laminar–turbulent transition experiment on the effect of surface imperfections on a natural laminar flow profile in compressible flow conditions

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Abstract

In the present study, the effect of various two-dimensional surface defects (forward-facing steps and ramps, backward-facing steps and ramps, gaps, and steps and gaps) on boundary layer transition was experimentally investigated in the compressible, subsonic regime. A laminar profile was specifically designed and manufactured by ONERA to allow for a maximum number of defects to be tested simultaneously, and to include resin pockets to accurately monitor laminar–turbulent transition using infrared thermography. Transition was also characterized using the ΔN model based on linear stability calculations. Relatively good agreement with existing ΔN models for forward-facing steps as well as gaps was found, indicating that these models, which were mostly developed for incompressible flows, can still be used as an initial estimate for compressible flows. One particular case of interest included a critical step and gap (for which transition occurred immediately downstream of the defect) where neither the gap nor the step component could be identified as mainly responsible for triggering transition. Steps and gaps should therefore be included whenever possible to the canonical shapes of defects investigated in transition experiments to further refine the different types of defect encountered in industrial application, and provide appropriate criteria for their allowable tolerances.

1 Introduction

As aviation focuses on reducing its emissions and fuel burn, one solution is to reduce the skin friction drag on the wings, nacelles, and empennage by delaying laminar–turbulent

transition. One effective approach consists in designing lifting surfaces with maximized extents of favorable pressure gradient, which tends to stabilize the boundary layer. This approach is referred to as natural laminar flow (NLF) and has already been successfully implemented in both wind tunnel and flight tests (cf. Holmes et al. 1984 for a comprehensive review of NLF experiments up to the 1980s); however, recent improvements in design and manufacturing techniques (Fujino 2005; Larsson 2022) allow to consider NLF wings as serious alternatives for commercial applications.

However, to assess commercial feasibility, further knowledge on the robustness of NLF wings to the presence of surface defects is necessary so that realistic manufacturing tolerances can be defined. In particular, two-dimensional surface defects are representative of the possible mismatch between panels of a wing, such as between the slats and main wing box. Both numerical and experimental investigations exist in the open literature on two-dimensional surface defects, such as backward- and forward-facing steps (BFS and FFS, respectively), but are often at low Reynolds numbers and on flat plates.

Various experimental and numerical studies were performed to quantify the effect of surface defects, which

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can broadly be characterized as either moving the transition position further upstream, with respect to the smooth case, or triggering transition at the location of the defect. In an initial study by Nenni and Gluyas (1966), critical Reynolds numbers, based on the dimensions of different types of defects, were provided as guidelines, but provided limited information on the range of validity of these criteria. Sinha et al. (1981) then performed some detailed wind tunnel experiments on characterizing the flow pattern and fluctuating velocity profiles downstream BFS for very low flow velocities (typically 1.8 ms^{-1}) and for large defect heights (up to $h/\delta_1 = 1.8$, with h the surface defect height and δ_1 the local boundary layer displacement thickness), while Perraud and Séraudie (2000) solved the two-dimensional incompressible Navier–Stokes equation assuming laminar flow around BFS and FFS. In both studies, the flow geometry around steps revealed the presence of small separation bubbles that destabilize the boundary layer by the introduction of an inflection-type instability. In particular, BFS exhibited greater bubble lengths and thereby had a more destabilizing effect on the boundary layer than FFS of corresponding height. Experiments by Wang and Gaster (2005) also showed that BFS tended to induce earlier transition than FFS, which has motivated more in-depth investigations on BFS in particular and their effect on transition depending on Mach number, Reynolds number and pressure gradient. Further experiments by Duncan et al. (2014a) and Duncan et al. (2014) provide yet more insight on the flow physics involved in the interactions between steps and the boundary layer using infrared thermography and hot-wire anemometry in both wind tunnel and flight environments. Numerical studies, such as by Edelmann and Rist (2015a), enable the investigation of the effects of FFS on boundary layer stability at different Mach numbers using DNS simulations and linear stability. Attempts to reduce the influence of steps by studying the effect of forward-facing ramps (FFR) on transition in flight test experiments on the X-21 and the T-34C are presented in Holmes et al. (1985). In this case, authors concluded by providing the same critical Reynolds number based on height for both FFS and FFR. However, numerical investigations performed by Zahn and Rist (2017) found that for an FFS with dimensions such that boundary layer transition was triggered immediately downstream of the step, an equivalent-height FFR triggered transition further downstream of its location. Thus, FFR were less destabilizing than equivalent FFS.

To model the effect of steps on boundary layer instabilities, Wang and Gaster (2005) experimentally studied the onset of transition for incompressible flows developing over a flat plate in the presence of both BFS and FFS. In this study, the upstream displacement of the transition location was correlated with a penalty on the N factor provided

by linear stability theory (LST) and now known as the ΔN model. A more detailed decomposition of the ΔN model into an N_{peak} value, representing a significant increase in the instabilities' amplification at the defect location, and a ΔN_{far} value that represents the amplification boost still perceived by the instabilities further downstream of the defect is presented by Béguet et al. (2016). In the present case, the reported ΔN values are assumed to correspond to the ΔN_{far} . In all the aforementioned studies, the presence of surface defects was always found to amplify the boundary layer instabilities responsible for transition and therefore move the transition position upstream. However, Wörner et al. (2003) performed a numerical study on the effect of BFS and FFS for incompressible flow over a flat plate. In this DNS study, some FFS cases were shown to stabilize the boundary layer modes responsible for transition.

The ΔN models were further extended by experiments by Crouch and Kosorygin (2020) for both FFS and BFS based on experiments, and validated by a numerical approach based on the linearized Navier–Stokes (LNS) equations performed by Hildebrand et al. (2022). Costantini et al. (2018) and Costantini et al. (2022a) performed an experimental study on the effect of FFS on the transition onset over non-adiabatic conditions and for compressible flows over a flat plate, respectively, to further investigate their influence on the resulting ΔN , while Heintz and Scholz (2023) also analyzed their BFS and FFS data using linear stability theory to calculate ΔN values based on their experiments, and found that for the range of step heights that were tested (less than or equal to the local boundary layer displacement thickness), the behavior of BFS was better captured by the ΔN model than that of FFS. More recently, an experimental investigation on the influence of BFS on a flat plate in various Mach numbers, Reynolds numbers and pressure gradients by Risius and Costantini (2025) found a threshold BFS with $h/\delta_1 < 0.15$ and Reynolds numbers based on its

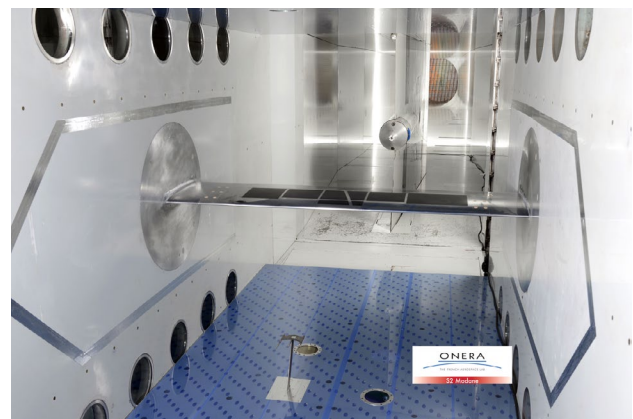
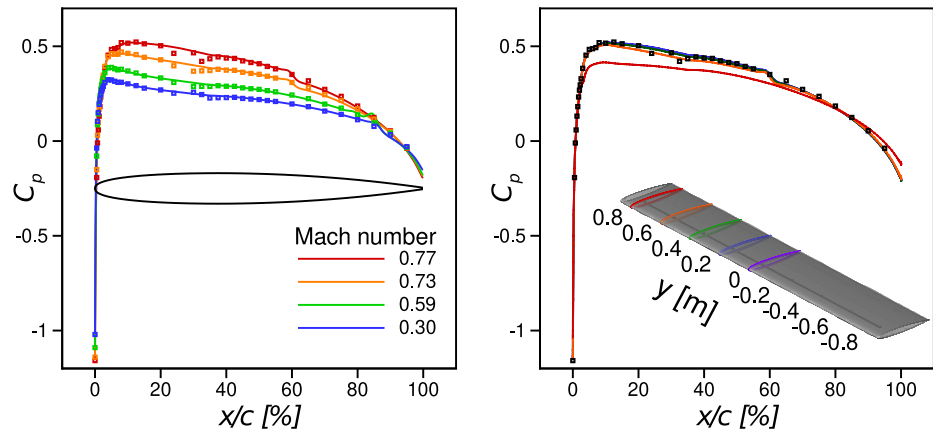


Fig. 1 Symmetric NLF profile model mounted inside the S2MA wind tunnel test section

Fig. 2 NLF profile: pressure coefficient (C_p) distribution (lines: C_p from CFD, dots: C_p from measurements) for several Mach number conditions and set to a zero lift angle of attack of the upper surface of the model overlaid with airfoil geometry



(a) Pressure coefficient for several Mach numbers (b) Pressure coefficient for several spans for $M = 0.77$

height $Re_h < 300$ under which the transition position did not move upstream, and the corresponding ΔN calculated using incompressible LST was slightly negative. Above this threshold though, BFS were found to reduce the extent of the laminar region, and the calculated ΔN were found to agree well with calculations from Hildebrand et al. (2020). Rouviere et al. (2023) proposed a neural model based on LNS calculation to model the effect on stability of steps and gaps in incompressible flow and found good agreement with recent experimental work from Hildebrand et al. (2022). Dimond et al. (2019) also proposed experimental work on step-and-gap configurations as well as gap configurations, while also investigating the effect of suction within the gap.

Many investigations on rectangular gaps have also been conducted and led to an initial classification in terms of their aspect ratio (ratio of depth to width) and the flow geometry inside their cavity (Charwat et al. 1961; Sarohia 1977; Sinha et al. 1982). Perraud et al. (2014) performed numerical studies to find that, in addition to aspect ratio, gap location and gap width had a significant influence on boundary layer instability amplification. Although initial results by Perraud et al. (2014) seemed to indicate that deep gaps should not have an effect on linear stability, Zahn and Rist (2015) found in their numerical study that boundary instabilities were further amplified due to resonance effects inside the gap. These results were then further validated by the experimental work of Risius et al. (2023). The combined effect of pressure gradient and gaps was investigated experimentally on an ONERA-D airfoil by Forte et al. (2015): In a strongly favorable pressure gradient, transition moved directly downstream of the gap, while in an adverse pressure gradient, the boundary layer was destabilized more

progressively and transition gradually moved upstream. Franco and Hein (2018) developed the adaptive harmonic linearized Navier–Stokes approach to further test gaps (and BFS) in more representative configurations, such as a transonic HLFC wing, and found that classic criteria (e.g., from Nenni and Gluyas (1966)) were strongly conservative, and could potentially be relaxed. In general, improvements in the prediction of the effect of surface defects on boundary layer transition are constantly explored to further expand the design and manufacturing possibilities of laminar flow airfoils.

Results from these studies are used to define either empirical criteria based on the defect dimensions and the local boundary layer thickness at the defect location (Nenni and Gluyas 1966; Béguet et al. 2016), or ΔN models (Crouch et al. 2006; Perraud et al. 2014; Crouch et al. 2022). However, given the complexity of laminar–turbulent transition experiments and simulations, most investigations are commonly performed in incompressible flow or at Reynolds numbers lower than those encountered by medium- to long-range aircraft. The purpose of the present study is therefore to provide extensive experimental data on the effect of surface defects on laminar–turbulent transition onset in compressible, subsonic flow, at Mach and Reynolds numbers closer to flight conditions, and compare these results to existing data.

In the present paper, the effect of FFS, BFS, FFR, backward-facing ramps (BFR), gaps and combinations of steps and gaps is investigated on a 2D NLF model in compressible, subsonic flow. The NLF airfoil was specifically designed for the present experiment so as to allow for a maximum number of surface defects to be

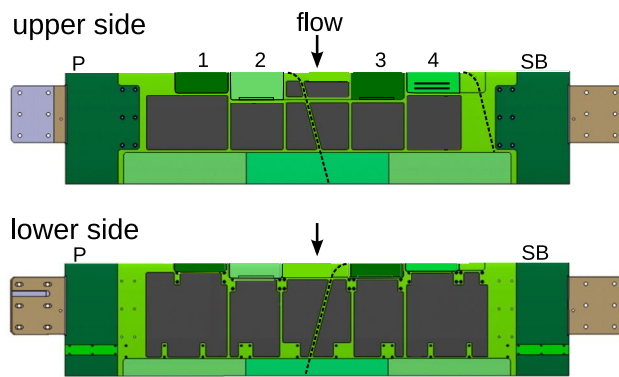


Fig. 3 Top and bottom views of the model with leading edge surface defect inserts, IR measurement regions and static pressure ports (exaggerated as the dashed black lines) (P: port side, SB: starboard side)

tested at once, as well as to include relevant instrumentation to study laminar–turbulent transition. First, the experimental method is presented, including the wind tunnel test facility, the NLF model, and the instrumentation. Next, the methodology used to determine the transition position using infrared thermography measurements is described. Finally, these measurements are analyzed for each type of surface defect and compared with existing data and models from the literature. Boundary layer analysis included performing Reynolds-averaged Navier–Stokes (RANS) computations for the base flow and calculations using linear stability theory to analyze the boundary layer stability and quantify the influence of surface defects on the transition process.

2 Experimental method

2.1 Experimental facility

Experiments were performed at the ONERA S2MA wind tunnel, a continuous pressurized flow facility with variable total pressure (to within ± 1000 Pa) and total temperature (to within ± 1 K). The Mach numbers 0.3, 0.59, 0.73 and 0.77 were investigated (± 0.001), therefore requiring the use of the $1.77 \text{ m} \times 1.75 \text{ m}$ (height $\times$

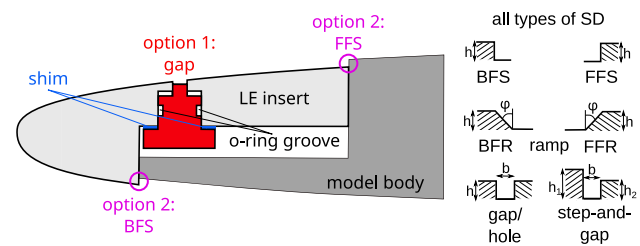


Fig. 4 Schematic of a leading edge insert with a single gap slot. In Option 1, the LE insert (in light gray) is flush-mounted with the model body, and shims (in blue) allow gap depth changes. In option 2, the gap slot is flush-mounted with the LE insert while the entire LE insert is shifted with respect to the model body, creating steps on the upper and lower surfaces. (Note: Options 1 and 2 were never tested simultaneously.) Also shown are canonical representation of the different types of tested surface defects (SD)

width) transonic test section (as opposed to the alternative supersonic test section). Nominally, the floor and ceiling of this test section are perforated; however, for the present campaign, both were taped to reduce acoustic disturbances and therefore provide flow conditions suitable for natural transition experiments. The flow quality was quantified by measuring the turbulence level in the test section using a hot-wire probe mounted at the floor of the wind tunnel, as illustrated in Fig. 1. A maximum turbulence intensity (evaluated over a frequency range between 3 Hz and 20 kHz) was always less than 0.20% for all Mach numbers, which corresponds to a threshold $N_{T,\text{Mack}}$ between 7–8 using Mack’s relation in Mack (1977). The turbulence intensity was measured at a single physical location. The turbulence intensity experienced by the model is not easily accessible in the S2MA wind tunnel. This $N_{T,\text{Mack}}$ value is provided for indicative purposes only: The smooth configurations for the different total pressure and Mach number combinations were used to evaluate the actual N_T .

In addition to varying the Mach number, a sweep in total pressure was also performed to allow the investigation of a wide range of Reynolds numbers. Total pressure $Pi0$ was therefore varied between 60 kPa and 130 kPa. With the range of tested Mach numbers, unit Reynolds numbers between $Re = U_\infty/\nu_\infty = 4e6 \text{ m}^{-1}$ and $22e6 \text{ m}^{-1}$ (to within 1 and 2%, respectively) were achieved.

Table 1 Leading insert characteristics: upper and lower side trailing edge position, and types of achievable surface defects

	Insert 1	Insert 2	Insert 3	Insert 4
Upper x_{TE} [% x/c]	19	25	25	19
Lower x_{TE} [% x/c]	7	13	13	7
Type of SD	BFS/FFS	BFS/FFS, S&G	BFS/FFS, ramps gaps ($x_G/c = 15$ & 22%)	BFS/FFS gaps ($x_G/c = 9$ & 15%)

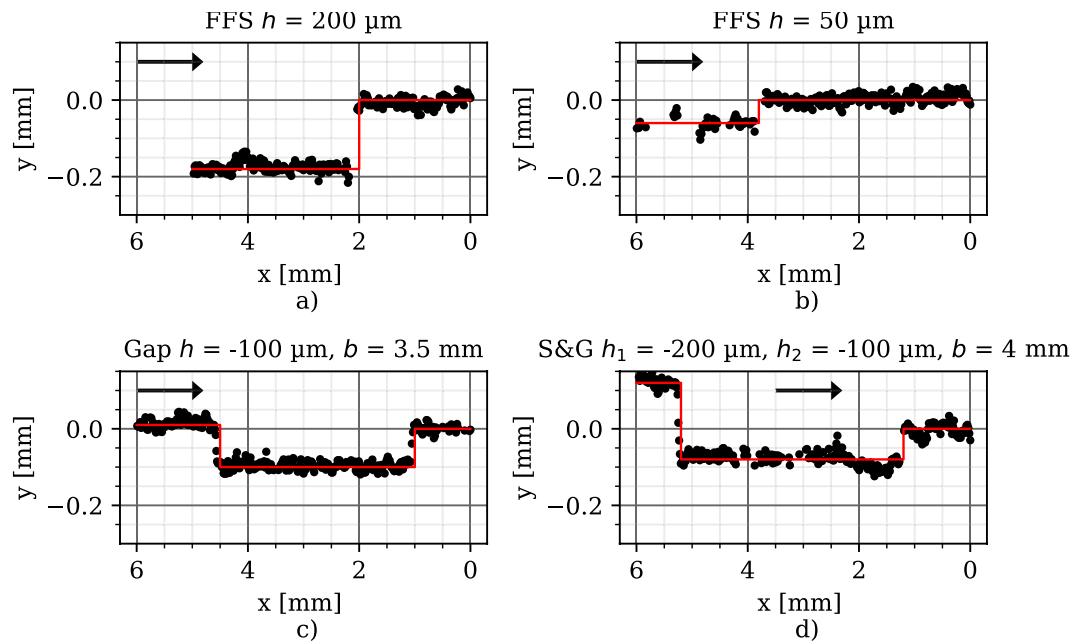


Fig. 5 Dimensions of defect on the upper surface evaluated using optical profilometry for configuration (all defects shown here were tested simultaneously). The red line shows the average positions for

2.2 Natural laminar flow profile geometry

A two-dimensional (2D) profile spanning 1.75 m, the entire width of the test section, was used during this investigation. The model was both designed and manufactured by ONERA in the framework of the French CORAC EPICE project, and is shown mounted in the test section in Fig. 1. The sting holder, seen in the background of the model picture, was present during testing but was located 2 m downstream of the model, thus avoiding any tonal acoustic disturbance. In order to comply with the objectives of a previous project, the profile geometry was designed by coupling an in-house inverse design optimizer to the CFD solver elsA (Cambier et al. 2013) to achieve a target pressure distribution representative of a nacelle pressure distribution that would guarantee a laminar boundary layer up to the shock location at the design Mach number equal to 0.8. Given the blockage constraints in the test section, which require a low relative thickness, the resulting airfoil is also representative of an airfoil for an NLF profile. In the present study, the maximum Mach number under investigation was always maintained at less than or equal to 0.77, which corresponds to the critical inflow Mach number value for this airfoil at zero angle of attack. By testing the profile at off-design Mach numbers, the pressure distributions show slightly adverse pressure gradients, as shown in Fig. 2a; however, the profile is still NLF due to the extended chordwise regions of laminar flow. The purpose of investigating this series of pressure distributions is to provide experimental data based on the geometry of a

each set of points (e.g., a single set being either the top or bottom of a step) and arrows show the direction of the flow

lifting surface relevant to industry applications, and compare them to existing transition models. The final profile geometry included a measured chord¹ c of 0.39648 m and a maximum thickness of $0.085c$ at x/c equal to 0.375.

Based on 3D CFD calculations around this profile for a Mach number of 0.77 (see Fig. 2b), the flow was considered two-dimensional in the spanwise direction over ± 400 mm from the centerline, thereby defining a region in which to include metallic inserts. Even up to $y = 0.6$ m, the pressure distribution is very similar to this central distribution. On these curves, the discontinuity around $x/c = 60\%$ is due to the onset of transition in the RANS computations. The influence of the lateral walls of the test section only becomes visible at $y = 0.8$ m, where the pressure distribution appears clearly different from the others. This wide extent of 2D flow enabled the design of four removable metallic inserts in which different surface imperfections could be introduced at the same time.

An overview of the upper and lower sides of the model is shown in Fig. 3. In order to facilitate comparisons between the upper and lower surfaces, the surfaces are projected along the same normal vector (with respect to the upper surface). As can be seen in Fig. 3, there is a direct match for each insert on the upper and lower surface. This standard is also used for IR images. The leading edge inserts are numbered 1 through 4 starting from the port side of the model. The wing model was mounted wall to wall in the test section

¹ The trailing edge is slightly truncated to prevent personnel injuries during handling and to avoid damage.

so that the widest spanwise extent of the model could be considered two-dimensional. Two lines of static pressure ports along the entire model chord are included on the upper side, located at the center line (42 ports) and on the starboard side (21 ports), to check for any deviations from the two-dimensional flow assumption. Additionally, another line of 21 static pressure ports is located on the centerline of the lower side to adjust the model's angle of attack at zero lift angle from an aerodynamic point of view, which is set for each Mach number, and does not correspond to the geometric zero angle of attack. This adjustment is validated once the pressure distribution is symmetric.

One of the main reasons for designing a two-dimensional model was to allow for a maximum number of surface defects (along the spanwise direction) to be tested simultaneously. All surface defect configurations were also tested in the zero lift angle of attack so that both the upper and lower surfaces could also be used at the same time. Each insert was 185 mm wide (in the spanwise direction) and could be independently removed and modified. Figure 3 shows one of the possible combination of surface defects, although different inserts from those shown in the figure were available for locations 2 and 4.

Table 1 summarizes the inserts' trailing edge chordwise positions depending on their spanwise position and the model surface, and lists the different possible types of surface defects that could be tested. As a note, when inserts were mounted as BFS or FFS, the other types of surface defects available (e.g., gaps) on that insert were positioned in their smooth configuration. Additionally, because the steps were set by placing precision ground shims to shift any given insert, for a BFS on the upper side, an equivalent-height FFS resulted on the lower side, albeit at a different chordwise location since the inserts' trailing edge positions varied between the upper and lower surfaces.

For inserts that include two gaps, only one gap was tested at a time. Figure 4 shows a close-up view of an insert with only one gap on the upper side. Combining O-rings and a new seal at each mounting provided an airtight seal at the interface between each surface defect and its corresponding insert, as well as between each insert and the model to prevent any recirculation that could interfere with the transition process being studied. Similar to how BFS or FFS were shifted vertically, the gap depth could be modified by shifting the gap slot insert (shown in red in Fig. 4) with shims. The different types of tested surface defects included: BFS/FFS, gaps, ramps, and a combination of a step and a gap, referred to as step and gap or S&G. Both ramps and steps and gaps were included to test more realistic surface defects, compared to canonical FFS/BFS or gaps. More specifically, and as shown in the schematic representation of all the different types of surface defects in Fig. 4, a step and gap is a gap having sides

of uneven height. For this type of defect, two heights, h_1 and h_2 , have to be specified along with the gap width b . All defects had a minimum width of 100 mm in the spanwise direction to satisfy the two-dimensional shape assumption, with FFS and BFS being wider. All defect dimensions were verified at each assembly using an LMI Gocator 2320 optical profilometer (height resolution between 1.8 and 3 μm). An example of the verification measurements is shown in Fig. 5 for all four defects of a single configuration on the upper surface. Each figure's subtitle shows the theoretical defect dimensions for comparison with the measurements. In a limited area around the defect (2 mm upstream and downstream 2 mm), a discrepancy of about 5 μm mean value is measured between the theoretical value of the defect and the actual measurement. Given that measurements and nominal dimensions were found to be within 5 μm of each other, the theoretical dimensions are therefore used for the remainder of this paper.

The ramp geometries were obtained by first shifting the metallic leading edge inserts with a shim by a height h , in the same manner as for steps. On the upper surface, the ramp's (streamwise) width was set to 3.5 mm and started at 24.1% of chord. On the lower surface, the ramp had a width was equal to 4 mm and started at 12% of chord. Depending on the location of the ramp (upper or lower surface), the slope of the ramp accounted for the local curvature of the profile. Therefore, given that the ramp width was fixed, the slope of each ramp changed for each tested height. On the upper surface, ramp slopes were between $\phi = 5.6^\circ$ and 11.2° , whereas on the lower surface, slopes were between $\phi = 1.1^\circ$ and $\phi = 6.1^\circ$.

2.3 Instrumentation

As mentioned previously, a total of three static pressure port lines extending over the entire chord length are available on the model: two on the centerline on the upper and lower surfaces, and one on the starboard side of the upper surface. Static pressure ports had 0.2 mm diameters. Static pressures were acquired at a sampling frequency of 4 Hz for a duration of 4 s on PSI modules. The two static pressure distributions at the center line were used to determine the angle of attack for a zero lift coefficient for each Mach number. To modify and monitor the angle of attack, each side of the model was mounted with a motorized hydraulic cylinder and two KEYENCE optical sensors, placed upstream and downstream of the model pitch axis. In practice, the KEYENCE optical sensors measured the distance between the sensor and a sighting support parallel to the model's chord from which the angle of attack of the model could be determined with an angular error of linearity of less than 0.004° . Both sides were instrumented identically to check for any torsion of the model.

Table 2 Infrared camera resolution

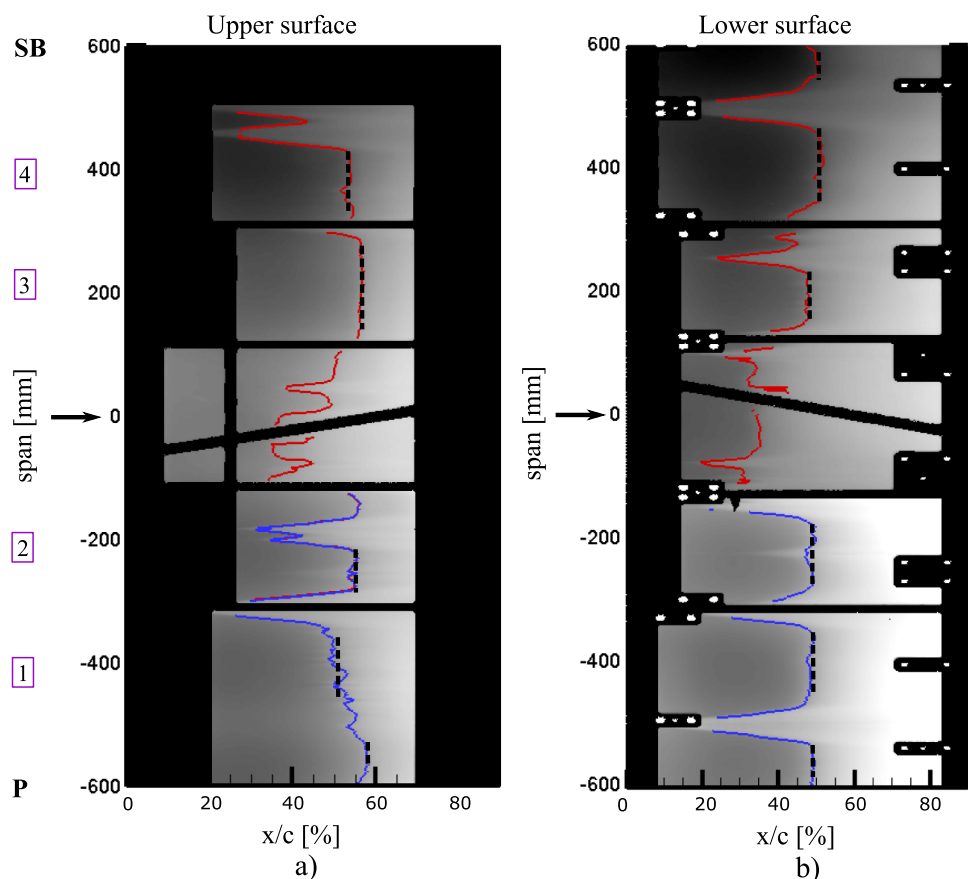
	FLIR SC5600	Infratec 9400	FLIR SC7750	IRC906
Resolution (mm)/pixel	1.35	0.45	1.35	0.8

Finally, the model included thermally insulated regions, shown in dark gray in Fig. 3, to enable the detection of laminar–turbulent transition using infrared (IR) thermography. A coating of resin was poured inside one-millimeter deep cavities in the model and subsequently machined to match the profile geometry. The resin offered a suitable compromise between low thermal conductivity (approximately $3 \text{ JK}^{-1} \text{ m}^{-1}$) and smooth surface finish (measured R_a of $1.14 \mu\text{m}$ [$\pm 0.005 \mu\text{m}$]), following recommendations from Le Sant et al. (2002), and well below the transition criterion in Smith and Clutter (1959). The working principle behind transition detection using IR thermography is the increase of the convective heat transfer and wall shear stress of a boundary layer in the turbulent regime compared that of the laminar regime. By generating a temperature difference between the ambient air and the model, the model wall temperature distribution will vary as a function of the local heat transfer, therefore reflecting the laminar or turbulent state of the

boundary layer. In the present case, the wind tunnel air temperature was modified by a maximum of 5 K (while keeping total pressure and Mach number constant) to create the temperature disequilibrium with the model without impacting the transition process. The impact was quantified by comparing repeatability runs throughout the entire test campaign: For a given combination of total pressure and Mach number, the total temperature variations to enhance contrast on IR cameras did not have a discernible effect on the position of the transition front detected on the IR cameras.

Four infrared thermography cameras (FLIR SC5600, Infratec ImageIR 9400, FLIR SC7750, and IRC906) were used to monitor simultaneously both upper and lower surfaces of the model at any given time. Two cameras were used to cover the entire spanwise extent of each surface. Each camera was mounted in a ventilated alcove set to atmospheric pressure and installed on a damping system to avoid perturbations from the wind tunnel vibrations. The cameras were calibrated with a heated black body before each run. Neon lights were used instead of the conventional projectors to avoid radiative transfer. The sampling frequency for the IR cameras was set to 4 Hz during 30 s to provide a mean temperature field over 120 images. Given the position of the cameras, their spatial resolution are available in Table 2.

Fig. 6 IR thermography temperature contours for the smooth reference (without defect) configuration at $M_0 = 0.77$, $Re = 7.9 \cdot 10^6$ with colored bounds based on a detected inflection point in the temperature gradient for both upper (a) and lower (b) surfaces and all surface defects. Darker gray regions correspond to laminar flow and the dashed black segments correspond to the mean (over the maximum spanwise extent) transition location for each insert



3 Transition position determination from IR measurements

The present section discusses how transition position was systematically extracted from the IR thermography measurements. Using data from configuration at a Mach number equal to 0.77, total pressure $Pi0$ at 60 kPa and total temperature $Ti0$ at 301 K, IR images were acquired using all four cameras. Example outputs both with and without defects are shown throughout the section.

3.1 Transition position in the smooth case

Figure 6a shows all resin inserts on the upper surface for the smooth configuration (i.e., without any defect). For this aerodynamic condition, dark gray regions correspond to regions of lower temperature, indicating laminar flow while light gray ones correspond to regions of higher temperature and, therefore, turbulent flow where convective heat transfer is enhanced. The slight change in contrast in the spanwise direction is an artifact of the instrumentation: Each surface is a combination of images from two IR cameras, each with their own sensitivity. The red and blue curves (each color corresponding to the field of vision of a camera) correspond to the location of the inflection point in the streamwise temperature distribution, which, in this context, is interpreted as the transition location (Costantini et al. 2021; Lalande et al. 2022). A more detailed description of the physics that justify this procedure can be found in Lalande et al. (2022).

For low-freestream-turbulence flows such as in the present study, the transition phenomenon actually corresponds to a region with streamwise extent, and thus, the definition of the transition position can vary depending on the measurement technique used. For example, when using hot-wire anemometry, the transition position often corresponds to the location of the first turbulent spot and therefore the beginning of the transition region (to within the resolution of the probing displacement). Presently, using IR thermography, the detected transition is closer to an intermediate location within the transition region, due to the thermal inertia of the resin inserts. However, the purpose of the present study is to discuss the relative shift in transition position for configurations with surface defects compared to their smooth reference. Therefore, the “absolute” value of the transition position is not as significant as ensuring a consistent approach across all configurations for transition detection, such as in the present case. Repeatability tests were performed throughout the test campaign on the smooth configuration over all inserts (representing over 300 cases) and were used to determine an uncertainty in the detected transition position of 3% of chord length.

Table 3 Smooth-case reference transition positions evaluated for all inserts on both the upper and lower surfaces, and expressed in percentage of total wing chord

	Insert 1	Insert 2	Insert 3	Insert 4	Average
Upper surface	50.9%	54.3%	56.6%	53.4%	53.8%
Lower surface	48.9%	49.4%	47.8%	50.9%	49.3%

In addition to the detected transition position, Fig. 6a also indicates the insert number (framed in purple). Similarly, Fig. 6b shows the measurements performed on the lower surface for the same configuration.

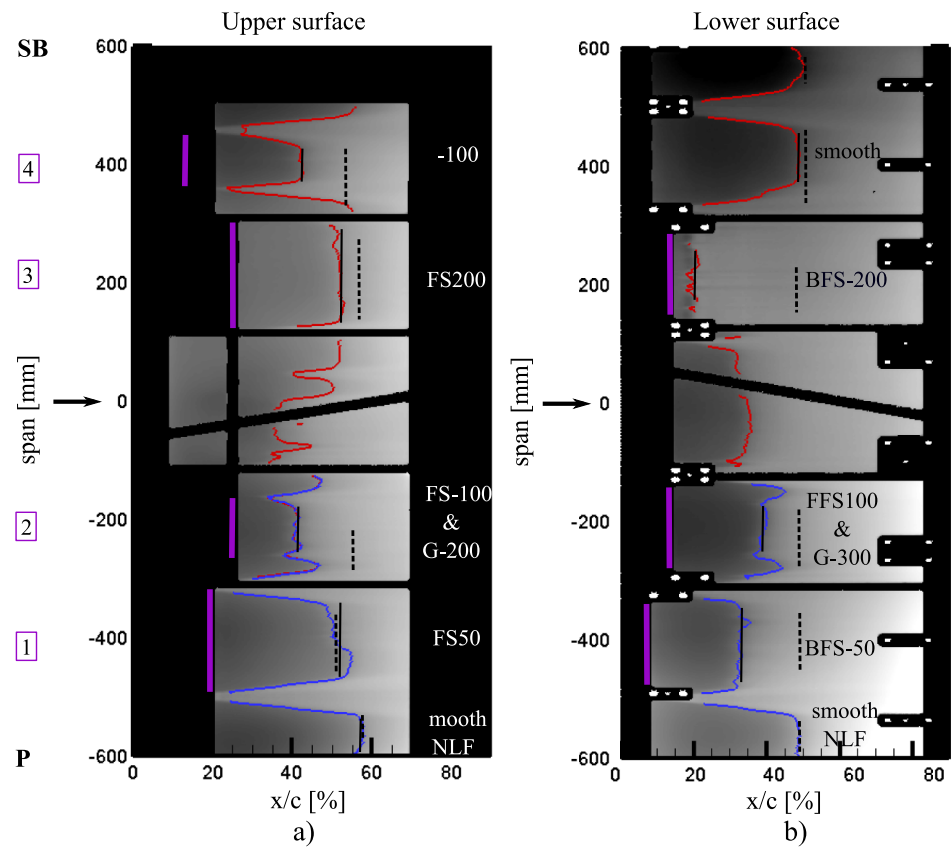
For this smooth configuration, the first observation is that the detected transition position at the centerline (spanwise y coordinates ± 100 mm) is uneven and furthest upstream compared to all other locations. In this centerline region, static pressure ports are distributed along the diagonal dark strip, inducing a discontinuity in the resin pocket which may destabilize the boundary layer and influence the transition onset. Additionally, a removable leading edge is present to allow access to the pressure ports. This combination of discontinuities is the most probable cause for the turbulent wedges. During the test, no intervention was able to improve this very localized issue.

The regions located in $-600 \text{ mm} < y < -500 \text{ mm}$ on both surfaces, were used as a smooth NLF reference position for real time interpretation of the results. The slightly oblique transition front on the upper surface between $-500 \text{ mm} < y < -450 \text{ mm}$ is due to the junction between insert 1 in its smooth configuration and the model. This region is therefore not included in the reference region. Note, however, that any subsequent data analysis was performed by considering each insert independently and using their corresponding smooth-case transition position as a reference.

Figure 6 also provides an overview of the flow’s two-dimensionality for what was considered the reference flow conditions: $M0 = 0.77$ and $Pi0 = 60 \text{ kPa}$. For each insert, the transition position is averaged over the widest continuous extent (e.g., $-280 \text{ mm} < y < -217 \text{ mm}$ for insert 2 on the upper surface, Fig. 6a). In this case, a summary of the transition positions is provided in Table 3. The upper surface has a transition position at approximately 54% of chord, compared to 49% on the lower surface so that a 5% (with respect to chord) difference in the average transition positions occurs between the upper and lower surfaces. A zero lift angle of attack was set, and pressure coefficient distribution between the two surfaces overlaps, excluding the possibility of an aerodynamic asymmetry.

However, one significant difference between both surfaces is the streamwise length of the leading edge inserts, compared to the streamwise length of the resin inserts. On the upper surface, the mirror-finish stainless steel leading

Fig. 7 IR thermography temperature contours for configuration at $M0 = 0.77$, $Re = 7.9 \cdot 10^6$ with gradient-based bounds of the laminar flow regions for both upper (a) and lower (b) surfaces and all surface defects (shown as the purple segments). The black (resp. dashed black) segments correspond to the mean transition location, over the maximum spanwise extent, for each insert with the surface defect (resp. in the corresponding smooth configuration, also shown in Fig. 7)



edge inserts extend to either 19% or 25% of chord before reaching the resin inserts. On the lower surface, the mirror-finish stainless steel inserts only extend down to 7% or 13% of chord, resulting in much longer resin inserts. Despite machining the resin to minimize the mismatch at the junction with the model's body, the resin has greater roughness than the mirror-finish stainless steel inserts.

In particular, the measured mean roughness R_a for the mirror-polish stainless steel was $0.33 \mu\text{m}$, which is then followed by the IR resin region with a measured R_a of $1.14 \mu\text{m}$, as mentioned previously. As early as in Smith and Clutter (1959), increased distributed roughness was found to affect boundary layer transition by moving the transition position closer to the leading edge. Therefore, the boundary layer on the lower surface, being in contact with the relatively rougher resin for a longer streamwise extent of 12% of chord, has a transition position closer to the leading edge, compared with the upper surface.

Focusing on the upper surface in Fig. 6a, differences in transition position between the outer inserts 1 and 4, and the inner inserts 2 and 3 can be observed. Once again, this difference can be attributed to the 6% of chord variation in the streamwise length between the outer inserts, starting at 19% chord, and the inner inserts, starting at 25% chord as summarized in Table 1.

The effect of the resin's distributed roughness is removed by the fact that, for each insert location, the corresponding smooth case is used as a reference. Since surface defects were introduced in the region of linear amplification of the TS waves, comparing their effect to the corresponding smooth reference allows to determine modifications to the TS amplification due to that defect. The difference between the defect and the smooth cases thereby isolates the contribution of the defect only, since in both cases, the contribution from the surface roughness is the same. Additionally, introducing surface defects in the region of linear amplification allows to compare results between all inserts, regardless of their chordwise position, because their effect is an addition (or subtraction) in amplification, as will be further discussed in the Results section on the ΔN model.

3.2 Transition measurement with surface defects

Figure 7 shows the similar temperature contours but from a configuration with surface defects. In this case, the location of the surface defects is shown (solid purple lines) to provide a reference of the spanwise extent and streamwise distance with respect to the leading edge of the resin insert.

Using the IR measurements in Fig. 7, the boundary defined by the temperature gradient can be further processed to determine the transition position that corresponds to a

particular surface defect. As mentioned previously, the entire leading edge inserts were shifted using shims to achieve FFS or BFS. As shown by the purple solid lines for inserts 1 and 3 in Fig. 7a, these defects span the entire widths of the inserts. Consequently, the region over which the transition position is evaluated maximizes the spanwise extent of the inserts. On the other hand, inserts 2 and 4 include centered gaps spanning only half of the inserts' width. For this reason, the transition front is only evaluated over the span shown by the solid black lines. As a reminder, labels with the type and dimensions of the surface defects are also provided on the figure.

A reference area labeled "smooth NLF" is also present on all IR measurements and, as expected, has the furthest downstream transition position compared to cases with defects. This smooth NLF sections on the model's port side were therefore used as sanity checks during real time comparisons.

However, for post-processing, each insert's corresponding smooth-case transition position was used as a reference when comparing with the configurations with surface defects. As explained in the previous subsection, this approach was used because of the slight variations in transition position that were observed between the upper and lower surfaces of the model, as well as the between the outer and inner inserts on the upper surface (where the majority of defects were inserted).

4 Effect of surface imperfections on laminar–turbulent transition

Using the methodology presented above, transition positions were extracted from the IR thermography measurements for all flow conditions and all surface defects. The present section synthesizes these results to compare with data published in the open literature and to establish the validity of existing criteria to the compressible flow regime. First, the approach to characterize this effect (in part based on LST) is described. Then, the results are divided into two subsections, based on the type of defects: The first subsection focuses on ramps, forward- and backward-facing steps (FFS and BFS, respectively); and the second subsection on gaps and steps and gaps.

4.1 Approach to characterizing transition with surface imperfections

As mentioned in the previous section, each insert was studied independently. The smooth transition position of each insert at given flow conditions (Mach and Reynolds numbers) was therefore used as the reference position.

The first and simplest characterization of the effect of surface defects on transition is to define a relative transition position, using the Δx_T parameter, previously used in Methel et al. (2019) (and defined in Crouch et al. (2022) as ξ), that allows easier comparison between experimental setups:

$$\Delta x_T = \frac{x_{T,SD} - x_{SD}}{x_{T,noSD} - x_{SD}}, \quad (1)$$

where $x_{T,noSD}$ stands for the transition location for the smooth case (without defect) and $x_{T,SD}$ represents the transition location downstream of a surface defect located at x_{SD} . For defects with dimensions that tend to zero ($x_{T,SD} \rightarrow x_{T,noSD}$), Δx_T tends to 1, whereas in the presence of a critical surface defect ($x_{T,SD} \rightarrow x_{SD}$), i.e., where transition occurs immediately downstream of the defect, Δx_T is equal to 0. For any intermediate values of Δx_T , the transition position is located between the smooth-case transition point and the surface defect: In this case, the surface defect is considered subcritical. The Δx_T uncertainty is equivalent to the uncertainty in the transition positions and is equal to 0.03. (Uncertainties for x_{SD} are related to manufacturing tolerances and are therefore two orders of magnitude smaller than that of the transition position.)

The second characterization involves performing linear stability theory calculations. In low-freestream-turbulence environments, modal transition occurs, as defined by Reshotko (2008). During the initial receptivity stage, freestream disturbances penetrate the boundary layer and excite its primary modes. In two-dimensional flow, such as in the present case, these modes, also known as Tollmien–Schlichting (TS) instabilities, develop.

The TS instabilities initially undergo a phase of linear amplification that can satisfactorily be modeled using LST through the computation of the N factor. Past a threshold amplification, secondary instabilities appear, quickly leading to the three-dimensional breakdown of the boundary layer, and the appearance of the first turbulent spot, marking the onset of transition.

The region of linear amplification has a large enough streamwise extent to allow modeling. Linear stability theory can therefore be used to calculate the spatial amplification rate of the TS instabilities across a range of frequencies. Spatially integrating these amplification rates results in N factor curves for each frequency, from which a single maximum N factor envelope curve can be determined.

These calculations can be performed for the smooth configurations of a given insert to determine a reference $N_{T,smooth}$ value for a given aerodynamic conditions ($M0$, Re), which corresponds to the threshold value past which secondary instabilities appear and the breakdown to turbulence starts. Therefore, for given flow conditions, $N_{T,smooth}$ is assumed to be constant. Based on this assumption, introducing a surface

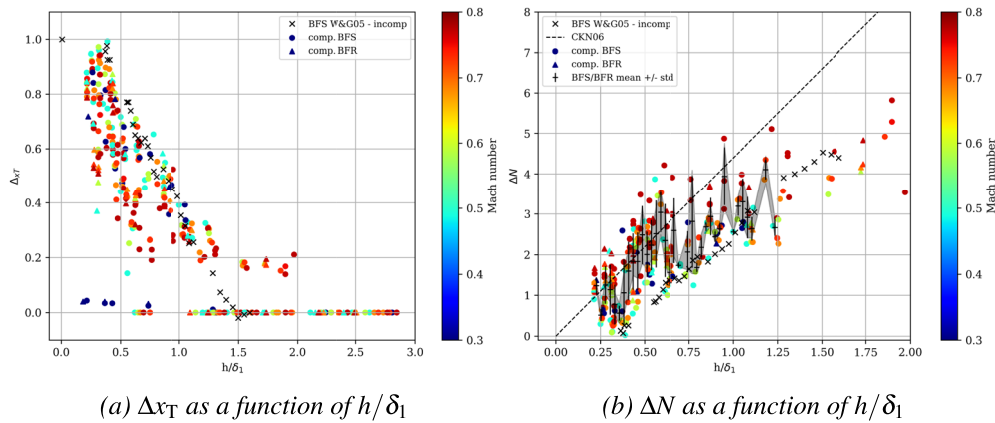


Fig. 8 Transition characterization as a function of h/δ_1 in terms of a) Δx_T and b) ΔN for BFR and BFS compared with Wang and Gaster (2005) (referred to as W&G05) and Crouch et al. (2006) (referred to as CKN06). Gray bands show the 95% confidence interval

defect that moves the transition position upstream implies that the threshold $N_{T,smooth}$ was reached earlier, and can be modeled by an amplification boost, ΔN , induced by the defect. The value of ΔN is defined as:

$$\Delta N = N_{T,smooth} - N_{x_{T,SD}} \quad (2)$$

where $N_{x_{T,SD}}$ corresponds to the N factor value at the transition position of the configuration with a surface defect. This method was successfully used on various types of surface defects (FFS, BFS and gaps) in studies such as Edelmann and Rist (2015b) or Crouch et al. (2006) and Crouch et al. (2022). More details on the ΔN model can be found in Methel et al. (2022).

Modeling the effect of surface defects using the ΔN model is a simple and accessible approach that cannot accurately capture the local flow structures, such as separation bubbles. By using LST to analyze the present data, the aim was: first, to provide experimental data in the subsonic compressible regime and compare them to existing data available in the open literature (also usually in the subsonic incompressible regime); and second, to use practical tools that can be used relatively easily in an industrial setting. A more detailed investigation of the flow around steps and gaps could be performed using either Parabolized Stability Equations or LNS to account for upstream effects of the locally rapid variations in boundary layer profiles on boundary layer stability; however, in the context of the present investigation (over 1000 test cases), and to remain within the bounds of industry applications, such approaches were not used.

In practical terms, the procedure for evaluating the ΔN values consists in first performing boundary layer calculations using ONERA's in-house code 3C3D. In the context of LST calculations, numerical pressure coefficient C_p distribution were extracted from steady RANS calculations using elsA at the spanwise location corresponding to the centerline

of the considered insert. The computational domain for the RANS calculations included the wind tunnel walls from the upstream contraction section down to the test section, where the wing model was mounted wall to wall, identical to the experimental configuration. The wall-resolved structured mesh included 10 million cells. Experimental values for total pressure and total temperature were applied. The mass flow in the outflow section was monitored so as to ensure the static pressure measured at a wall location upstream of the model in the wind tunnel was reproduced at the same location in the simulation with a precision below 0.5%. This criterion led to very good agreement between experimental and numerical C_p values, especially in the suction peak region as shown in Fig. 2. Numerical C_p distributions were preferred over experimental ones because of their greater smoothness, thus avoiding unphysical boundary layer separation in the boundary layer solver 3C3D. Next, using the high-resolution boundary layer velocity profiles from 3C3D, LST calculations were then performed using ONERA's in-house code CASTET, which solves the Orr–Sommerfeld equation for compressible flow.

Using the resulting envelope N factor distribution for each flow condition, the smooth $N_{T,smooth}$ and corresponding $N_{x_{T,SD}}$ for each defect configuration were evaluated and are discussed in the following sections. The study presented here represents more than 1000 post-processed configurations (defect type, defect size, total pressure, Mach number). An uncertainty of ± 0.2 on the N factors was evaluated and $\pm 2 \mu\text{m}$ for the displacement thickness δ_1 in the regions where surface defects were located. Based on this latter value and the geometric uncertainty for the surface defects, the uncertainty on the relative dimensions h/δ_1 was found to be ± 0.1 . The overall procedure is schematically presented in Appendix 7 along with a short description of the methodology applied to assess the uncertainties on the N factor values.

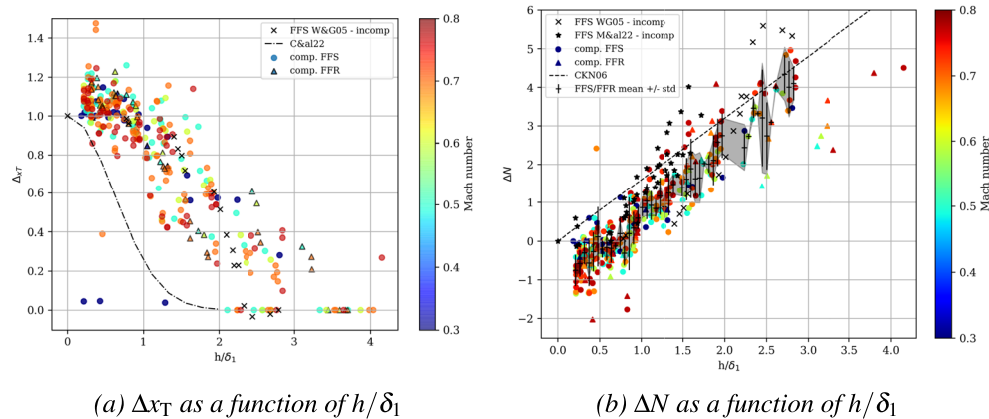


Fig. 9 Transition characterization as a function of h/δ_1 in terms of (a) Δx_T and (b) ΔN for FFR and FFS compared with Wang and Gaster (2005) (referred to as W&G05), Crouch et al. (2006) (referred to as

CKN06), Methel et al. (2022) (referred to as M&al22) and Costantini et al. (2022a) (referred to as C&al22). Gray bands show the 95% confidence interval

4.2 Backward-facing steps and ramps

The effects of steps and ramps were studied in terms of the evolution of the Δx_T parameter and ΔN as a function of the non-dimensional height h with respect to the local boundary layer displacement thickness δ_1 in the corresponding smooth configuration. This displacement thickness was calculated with the boundary layer solver 3C3D.

Figure 8a shows the results for BFS and BFR. The first observation is that, as a general trend, increasing h/δ_1 moves the transition further upstream, regardless of the type of defect. No significant difference between the BFS and BFR can be observed in terms of the critical h/δ_1 , where Δx_T starts to consistently be equal to zero for $h/\delta_1 > 1.1$ in Fig. 8a. The flat plate experiment results for incompressible

flow by Wang and Gaster (2005) are also plotted in this figure (black crosses) and exhibit a higher critical height around $h/\delta_1 \approx 1.5$. The second general observation is that the present data do not reveal an influence of the Mach number, similar to results found in Costantini et al. (2022a).

Next, Fig. 8b shows the evolution of the ΔN as a function of h/δ_1 . The information in this figure is complementary to that in Fig. 8a: As the non-dimensional step or ramp height increases, the decrease in Δx_T (representing the upstream movement of the transition position) is also related to an increased amplification boost ΔN . Note that all critical defects, which have a Δx_T value of 0, are not shown in Fig. 8b. For such defects, the basic assumption behind the ΔN model might not be valid, since critical defects might introduce more than a simple amplification

Fig. 10 Extracted measurements from IR contours for flow conditions $M0 = 0.683$, $Pi0 = 60$ kPa, $Ti0 = 300$ K for corresponding smooth, FFS and BFS with $h = 50$ μm at the same streamwise location

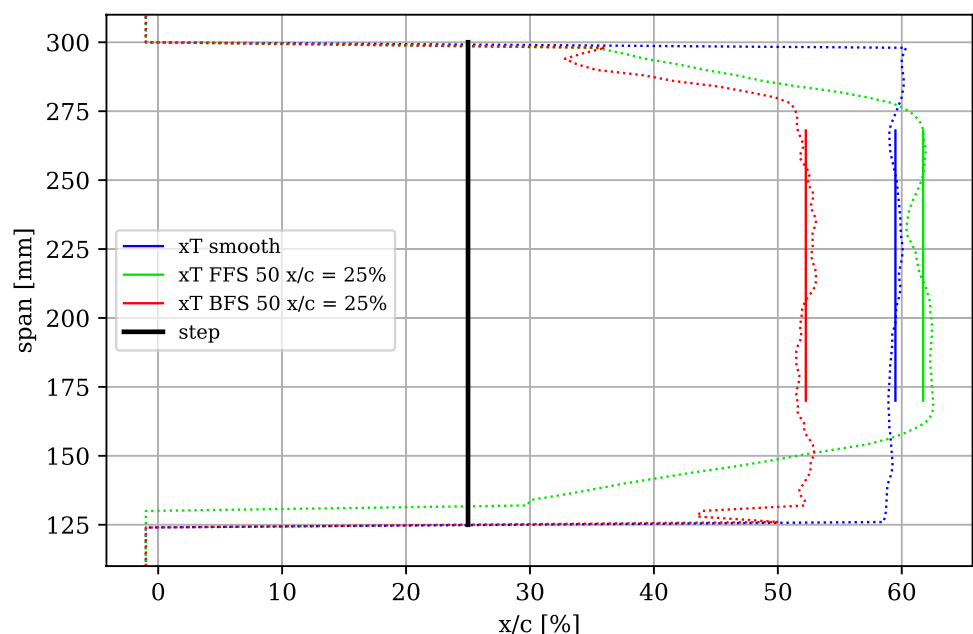
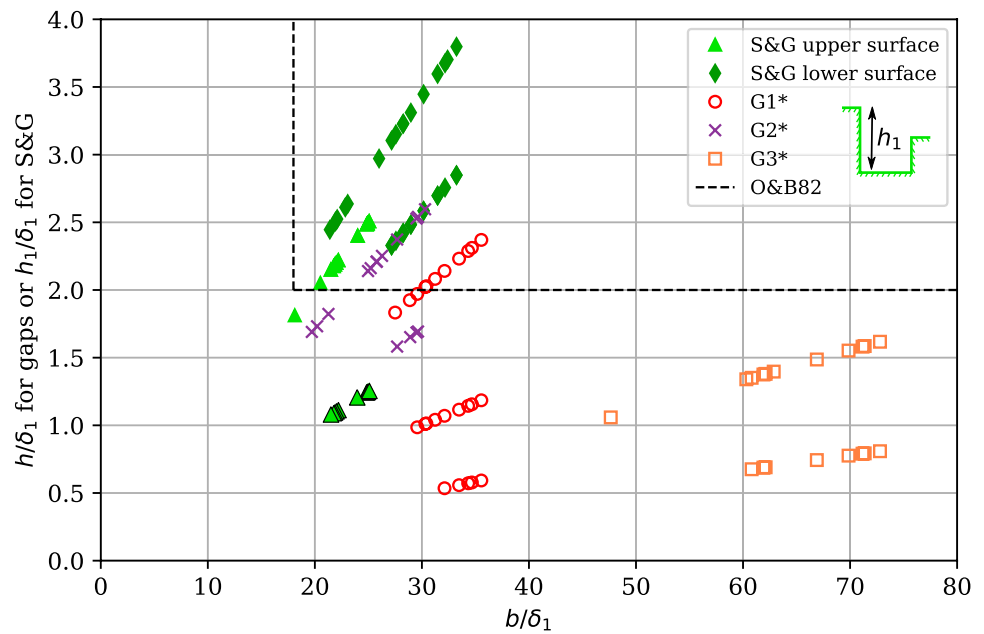


Fig. 11 Comparison of Olive and Blanchard (1982) criteria with critical steps and gaps. Gaps G1, G2, and G3 are shown with an asterisk since their corresponding transition position could be located between the leading edge of the IR insert and the surface defect, i.e., not necessarily critical. Green triangles with a black outline correspond to the unusual step-and-gap cases discussed in Sect. 4.5



boost to existing boundary layer instabilities. In particular, critical surface defects can generate inflection point instabilities that cannot be modeled using the present approach, which assumes that a surface defect enhances boundary layer instabilities already present upstream of the defect. For this reason, critical defects of all types will be excluded from any figure involving the ΔN in the remainder of the paper.

The ΔN values calculated for the BFR and BFS show some agreement with the linear ΔN model based on incompressible experiments from Crouch et al. (2006), shown as the black dashed lined labeled CKN06 in Fig. 8b. In particular, the general slope across all data points can be approximated linearly; however, the CKN06 model does not encompass some of the highly disruptive effects of defect with $h/\delta_1 < 1$. The scatter exhibited by the present data is

quantified by the 95% confidence interval calculated for various BFS dimensions, and shown as the gray band in Fig. 8b. The scatter is significantly greater than the ΔN uncertainty, indicating that ΔN models for BFS could be complemented by bounds to encompass this variability. Additionally, no differences in the ΔN values for BFS and BFR are observed; however, this agreement could mainly be due to the particular ramp slopes presently tested. The ramp is 4 mm wide and several slopes are tested [0.7°, 1.4°, 2.9°, 5.7°]. Further investigation into ramp geometry should be considered before confirming this finding.

4.3 Forward-facing steps and ramps

Similar comparisons were made between forward-facing ramps (FFR) and forward-facing steps (FFS), as shown in

Fig. 12 Comparison of the ΔN as function of h/δ_1 for all gaps G1, G2, and G3 with the gap ΔN model (dashed lines) proposed by Crouch et al. (2022) and the BFS ΔN model from Crouch et al. (2006)

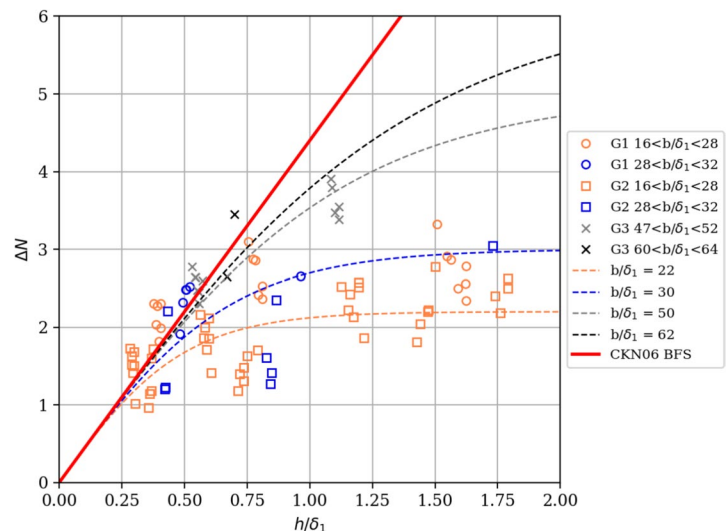


Fig. 13 Influence of S&G on transition in terms of Δx_T and ΔN as a function of h/δ_1 . Comparison to BFS (Fig. 8)

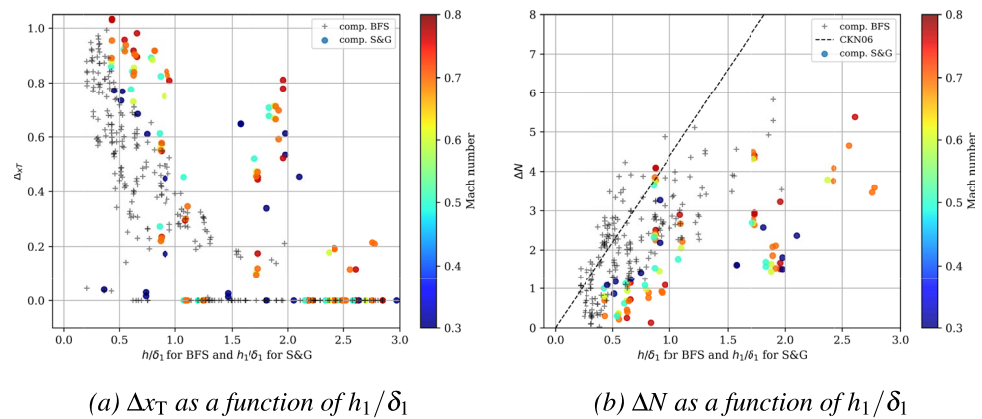


Fig. 14 Influence of S&G on transition in terms of Δx_T and ΔN as a function of h/δ_1 . Comparison to BFS (Fig. 8)

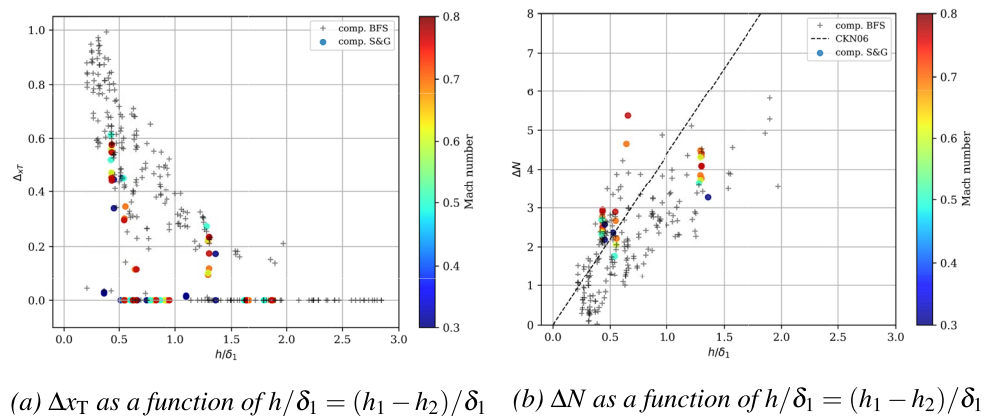


Fig. 9a. The forward-facing ramps are 3.5 mm wide, and several slopes are tested [0.8° , 1.6° , 3.3° , 6.5°]. Once again, the general trend is that increasing h/δ_1 moves the transition position closer to the defect location (decreasing Δx_T) by inducing greater ΔN , for both steps and ramps. Additionally, the variability of the FFS and FFR data is shown as the gray band in Fig. 9b, indicating the 95% confidence interval; however, as opposed to the BFS and BFR, despite the variability, the Crouch et al. (2006) linear model from subsonic data encompasses most of the present data. As for BFS, in the subcritical regime where $0 < \Delta x_T < 1$, the forward-facing ramps tested during this campaign were not found to be any less destabilizing than equivalent FFS, despite having mild slopes (always less than 11°). However, the difference between steps and ramps can be observed in terms of their critical dimensions: Critical FFS start to appear around h/δ_1 equal to 2, while critical FFR are closer to h/δ_1 equal to 3.5.

Compared to BFS, there is better agreement with the incompressible results from Wang and Gaster (2005), where the criterion was found to be $h/\delta_1 > 2.3$. One possible physical explanation could be that, given the greater destabilization caused by BFS, the sensitivity of the boundary layer stability could be further accrued by flow compressibility. In general, the present study in the compressible, subsonic

flow regime also confirms that BFS are more destabilizing than FFS. Recently, Costantini et al. (2022a) conducted a test campaign on a flat plate with FFS in the compressible regime. They found correlations (polynomial of degree 2) for each tested Mach number. In Fig. 9, the correlations obtained at $M_0 = 0.77$ from Costantini et al. (2022a) are plotted. There is relative agreement between this correlation and the presented experimental results. Nevertheless, for h/δ_1 ratios higher than 2, the trend is not consistent with the one shown by Costantini et al. (2022a), which could be explained by the fact that such defect sizes were not investigated during their study.

In Figs. 8 and 9, the data points are color-coded according to their corresponding upstream Mach number, ranging from 0.3 to 0.8. No specific trend related to the Mach number appears, similar to findings from Costantini et al. (2022a). When color-coding data points based on the Reynolds number, no specific trend was found either. Based on the present results on the impact of 2D steps and ramp defects on a particular transition mechanism (i.e., one governed by TS instabilities), compressibility does not seem to affect the critical dimensions for which the transition moves up to the defect location.

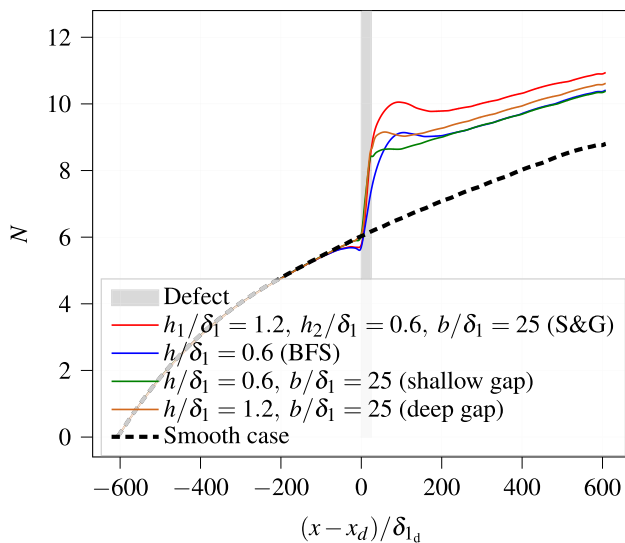


Fig. 15 N factor through incompressible LNS simulation on a flat plate for a S&G and all different combinations of corresponding BFS or gaps

One feature of particular interest in Fig. 9b is that some ΔN values are negative for FFS and FFR with $h/\delta_1 < 1$, corresponding to cases where the detected transition position with a surface defect is further downstream than in the smooth case. Some of the low- ΔN cases could be attributed to uncertainty in detecting the transition position; however, ΔN values close to -1.5 are hard to justify with a measurement uncertainty of ± 0.2 . Additionally, in the case of BFS, no aggregate of negative ΔN values were found. The negative ΔN seen for FFR and FFS could therefore indicate some slight stabilization, as illustrated in Fig. 10. In this figure, the transition positions with an FFS and BFS of identical h and h/δ_1 are compared to that of the smooth case: The BFS case is clearly destabilizing, whereas the FFS case has a transition position further downstream of the smooth case. Numerical simulations by Wörner et al. (2003) had previously shown that such stabilization from an FFS could occur with an h/δ_1 equal to 0.235, i.e., in the same range as in our experiments. In the same way, using RANS computation integrated to a transition model, Bhatia et al. (2021) showed the potential of a micro-FFS ($h/\delta_1 = 0.1$) to delay the laminar to turbulent transition and reduce drag. Incompressible flow calculations were also performed during the analysis of the present data and are shown in Appendix 8. Although no stabilizing effect was observed in this flow regime, the combination of an adverse pressure gradient followed by a favorable pressure gradient in the immediate vicinity of the step was found, and resulted in small ΔN values, with respect to the smooth case (in the order of 0.3). In accordance with the other works previously cited, the stabilizing effect of the FFS found experimentally in this paper could be due to the boundary layer

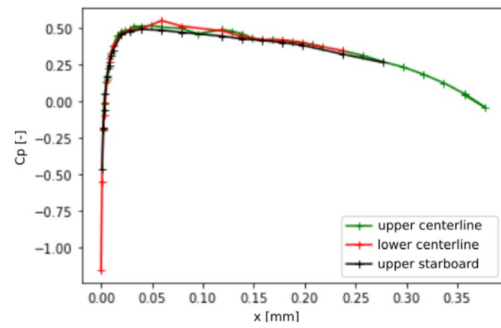


Fig. 16 Pressure coefficient distributions over all three pressure lines to show aerodynamic symmetry and two-dimensionality for $M = 0.77$

stabilization in the upstream region of the step outweighing destabilization in the downstream region. Although stabilizing a boundary layer by introducing an FFS cannot be considered a viable solution, such results provide supporting evidence that FFS, being generally less destabilizing and potentially stabilizing, are to be favored when considering laminarity, as opposed to BFS.

4.4 Gaps and steps and gaps

Next, gaps and steps and gaps (S&G) were analyzed and compared, starting with critical dimensions. Figure 11 shows the conjunct critical dimensions, height h/δ_1 and width b/δ_1 , compared to the existing criteria first established by Olive and Blanchard (1982) in incompressible flow, and published in Béguet et al. (2016). For the S&G, the depth that was used in the figure is h_1/δ_1 , which corresponds to the most upstream side of the gap. Olive and Blanchard criterion specifies that the gap will trigger the transition at its location if $h/\delta_1 > 2$ and $b/\delta_1 > 18$.

Gaps G1, G2, and G3 are shown with an asterisk on the figure's legend because their transition positions should be considered with caution. For example, gaps G1 and G2 are located on insert 4, as shown in Fig. 3, and only one gap at time was tested for a given configuration. However, regardless of the configuration, a non-negligible distance (between 3% and 9%) between the trailing edge of the gap and the leading edge of the resin insert exists. Therefore, there exists a region for which the transition position is unaccounted. Surface defects were flagged as critical once the transition position moved upstream of the resin insert's leading edge, leading to a possible underestimation of critical dimensions. These data are thus included in Fig. 11 but should be considered as indicative, and cannot rigorously be compared to the critical steps and gaps nor to the Olive and Blanchard (1982) data.

However, given their position on both the upper and lower surfaces (at the junction between the trailing edge of the metallic insert and the leading edge of the resin insert), S&G

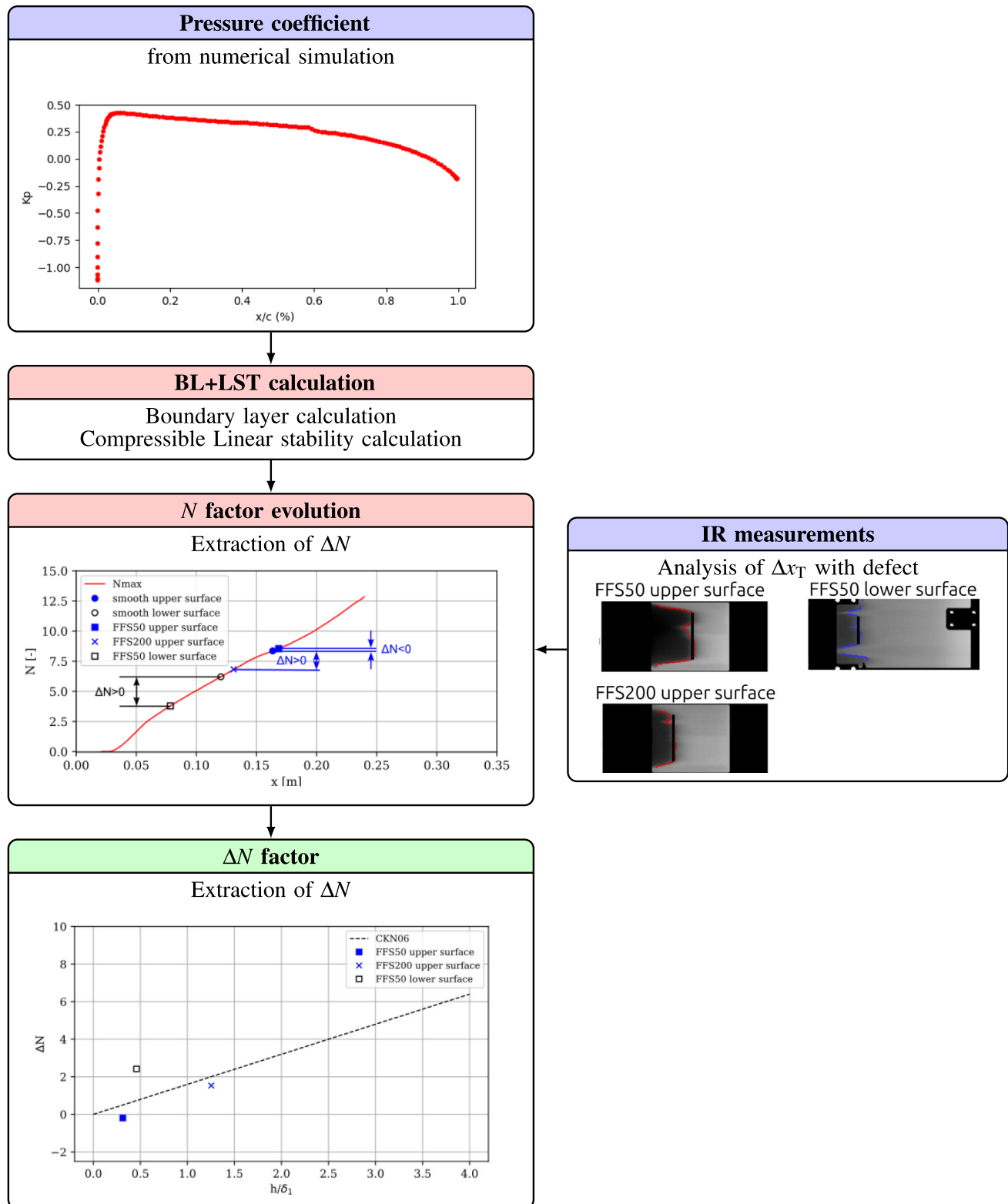


Fig. 17 Schematic procedure to determine the ΔN values for three surface defects located on the upper and lower surfaces of the same insert at $M_0 = 0.684$, $Re = 16 \cdot 10^6$

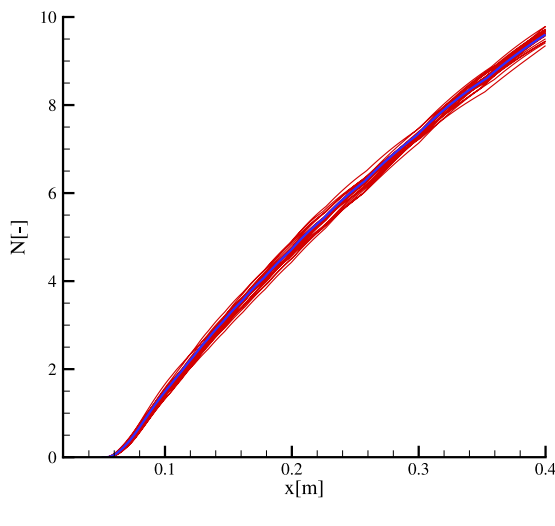


Fig. 18 N factor from 25 random pressure distribution (red lines) derived from the RANS pressure distribution (blue line) considering the standard deviation between experimental results and RANS results, $Mo = 0.684$, $Re = 16 \cdot 10^6$

identified as critical are reliable. Comparing the critical dimensions shown as diamonds and triangles to Olive and Blanchard (1982), good agreement is found for the majority of the cases considered. However, the critical S&G lying below the conjunct Olive and Blanchard criteria ($h/\delta_1 < 2$, $b/\delta_1 < 18$) can be considered in more detail before questioning the validity of the known criteria in the compressible regime. As mentioned previously, Fig. 11 characterizes S&G according to their dimension h_1/δ_1 , which, on the upper surface, corresponds to the largest depth possible. However, S&G can also be characterized by the most downstream side of the gap, i.e., h_2/δ_1 or by its step size h/δ_1 defined as the difference $h_1/\delta_1 - h_2/\delta_1$.

Next, the influence of subcritical defects is analyzed. The influence of canonical gaps on transition is shown in Fig. 12. Each type of symbol corresponds to a given geometric gap width, while colors correspond to the gaps' range of non-dimensional width with respect to the local boundary layer thickness, b/δ_1 . The dashed lines correspond to the ΔN model for gaps proposed by Crouch et al. (2022) that depends on both h/δ_1 and b/δ_1 according to the following expression:

$$\Delta N \approx 0.122 \frac{b}{\delta_1} \tanh\left(44 \frac{h}{b}\right) \quad (3)$$

Using this model, the dashed lines asymptotically tend toward a given ΔN value (depending on a given b/δ_1) as h/δ_1 increases. This leveling off of the curves is referred to as the "gap width limit" and corresponds to a deep gap ($b/h < 43$) behavior. The solid thick red line is the BFS ΔN model from

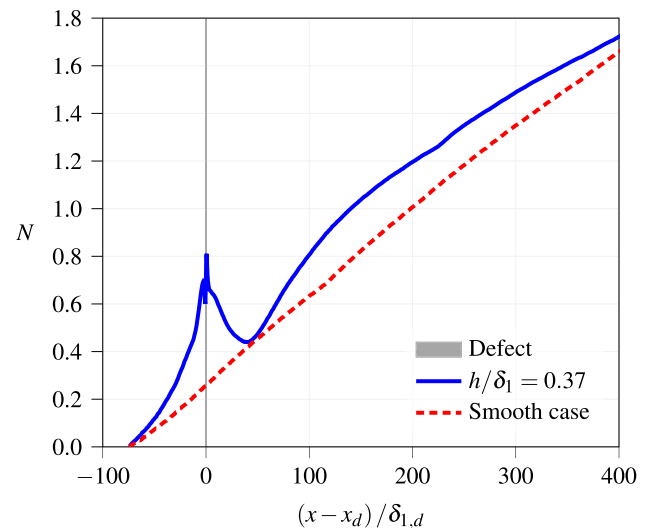


Fig. 19 LNS simulation on a flat plate for a FFS ($h/\delta_1 = 0.37$) and for a Reynolds number based on $\delta_{1,d}$, $Re_{\delta_1} = 550$

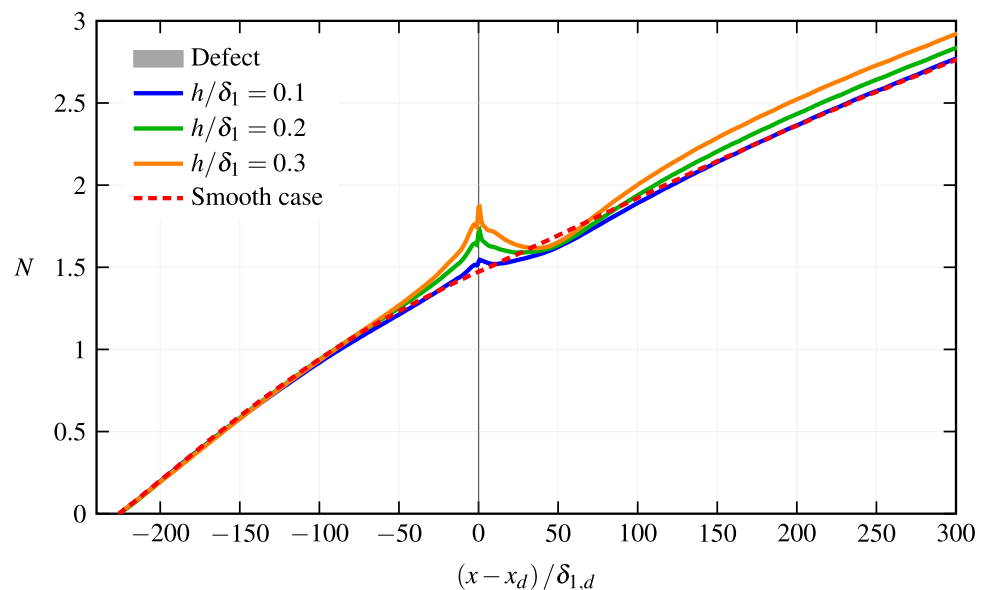
Crouch et al. (2006) that is used as the asymptotic limit to which gaps of extremely large width tend.

Compared to the data from the present experiment, the ΔN gap model slightly underestimates the effect of gaps: Especially in the case of the gaps with $16 < b/\delta_1 < 28$ (orange symbols), the ΔN values from the experiments are generally greater than those calculated by the model. However, only h/δ_1 up to 1.75 were tested in the present study, as opposed to values of 5 or 10 in the case of Crouch et al. (2022), where extremely deep gaps ($b/h \ll 43$) can be observed. Although the present data cannot be used as validation for the ΔN model in the compressible, subsonic regime, the beginning of a deep gap behavior (i.e., asymptotically tending toward a constant ΔN) seems to occur for the $16 < b/\delta_1 < 28$ range. Due to limitation in the wing model, deeper gaps were not tested but could be included in future work.

In the first part of this section, S&G were compared to gaps by considering their depth component; however, their depth and step components can also be compared to equivalent steps. Thus, for a S&G, the depth component can be considered as the upstream side of the S&G, referred to as h_1 in Fig. 4a, while the step component as the difference between its two sides, i.e., $h_1 - h_2$. Figure 13a shows the Δx_T values for the backward-facing S&G tested on the upper surface considering its normalized depth component h_1/δ_1 , while Fig. 14a considers the step component $h/\delta_1 = (h_1 - h_2)/\delta_1$. Both results are compared to the present study's BFS results that were presented earlier, in Fig. 8.

As the depth component h_1/δ_1 increases, S&G can be considered less destabilizing than the BFS, as indicated by the slower decrease in Δx_T (or slower increase in ΔN).

Fig. 20 LNS simulation on a flat plate for several FFS ($h/\delta_1 = 0.1, 0.2, 0.3$) and for a Reynolds number based on δ_1 , $Re_{\delta_1} = 1100$



However, an estimate for critical S&G would be h_1/δ_1 equal to 1.1 by choosing the first h_1/δ_1 for which Δx_T is equal to zero. This critical dimension is identical to the criterion for BFS, indicating that the depth component of S&G is mainly seen as a step by the boundary layer (for the gap widths presently tested, at least).

Next, when considering the step component h/δ_1 in Fig. 14, less spread between in the S&G data is observed. In this case, the step component reveals that the boundary layer is more sensitive to a backward-facing S&G than to an equivalent BFS, since ΔN values are generally greater or equal for backward-facing S&G than corresponding BFS. In particular, considering the cluster of critical S&G ($\Delta x_T = 0$), step dimensions for critical S&G would be determined in the vicinity of $0.5 < h/\delta_1 < 1$. For a majority of these cases, the depth component (in accordance with the conjunct Olive and Blanchard criteria) can be considered as mainly responsible for the critical nature of the defect. These results indicate that, for the most part, S&G can be considered as either driven mainly by their step or their depth component.

4.5 Unusual step-and-gap case

The S&G with dimensions around ($h_1/\delta_1 = 1.2$, $h_2/\delta_1 = 0.6$, $b/\delta_1 = 25$), shown in Fig. 11 as the green triangles with a black outline, is intriguing since neither a clear step nor depth effect can be particularly identified as the main contributor resulting in a critical surface defect. The conjunct parameters ($h_1/\delta_1 = 1.2$, $b/\delta_1 = 25$) are below the Olive and Blanchard criteria, while the step component $h/\delta_1 = h_1/\delta_1 - h_2/\delta_1 = 0.6$ is clearly outside the range of known BFS criterion. Before suggesting to modify the critical BFS criterion, a stability analysis of

this test case was performed to further understand this S&G combination.

Linearized Navier–Stokes calculations using harmonic forcing were performed using ONERA’s in-house code PIMS 2D for the S&G geometry mentioned above and the different combinations of the corresponding BFS or gaps on a flat plate. Flow conditions were set such that the Reynolds number based on displacement thickness, Re_{δ_1} was equal to 2300, which approximately corresponds to the case with $Pi0 = 100$ kPa, $Ti0 = 285$ K, and $M0 = 0.492$. Contrary to linear stability theory, the method used presently monitors the amplification of the TS instabilities by determining the maximum absolute value of the streamwise velocity fluctuations along the wall-normal direction, which is then normalized by a chosen initial amplitude. The amplification factor N is then determined by taking the logarithm of the amplification. Figure 15 shows the evolution of the N factors along the streamwise direction, described as the distance in the x -direction with respect to the surface defect and non-dimensionalized by the displacement thickness at the defect location, $\delta_{1,d}$. More details on the numerical method used to perform the stability calculations can be found in Rouviere et al. (2022).

From the stability calculations, the main striking observation is that the S&G reaches the largest N factor value, close to 10, compared to the smooth case or any of the other types of defects shown in Fig. 15. On the other hand, the shallow gap $h/\delta_1 = 0.6$ has the smallest amplification peak, while the deep gap $h/\delta_1 = 1.2$ and the BFS both reach slightly greater N factors, at approximately 9. This combination of S&G therefore seems to be more destabilizing than any single defect based on one of the S&G’s geometric component. Although these cases (identified as the green triangles lying

far below the Olive and Blanchard criteria in Fig. 11) are few and clustered around the same non-dimensional geometric parameters in the present data set, these results demonstrate that S&G cannot be considered as a step or a gap, and its specific geometry has to be considered as a canonical surface defect shapes.

5 Conclusion

An experimental investigation was performed on the effect of surface defects (BFS, FFS, ramps, gaps and steps and gaps) on the laminar–turbulent transition of a laminar profile in compressible, subsonic flow at the ONERA S2MA wind tunnel facility. The two-dimensional model was set to a zero lift angle of attack, such that the four leading edge inserts (in which various surface defects of controlled dimensions could be introduced) enabled the acquisition of transition data on both upper and lower surfaces simultaneously.

The model was also equipped with resin pockets located downstream of the metallic leading edge inserts so that laminar–turbulent transition could accurately be monitored in real time using infrared thermography. All test cases were analyzed with the same methodology based on determining the location of the inflection point in the streamwise temperature gradient. Additionally, the slight differences in streamwise lengths and average roughness of the mirror-polish metallic inserts and resin inserts led to slight variations (to within 5% of the chord) in the smooth-case transition position. For both these reasons, only the difference, as opposed to the absolute value, in transition positions or N factors between the surface defect case and the corresponding smooth case was analyzed.

The non-dimensional difference in transition positions was evaluated through the Δx_T parameter that is equal to 1 when the transition position corresponds to the smooth case and 0 in the critical transition scenario, where transition occurs immediately downstream of the surface defect. Comparing the present FFS data to results from incompressible flow experiments on flat plates, the existing empirical criteria are still valid in compressible flow, with critical $h/\delta_1 > 2.3$. Although subcritical FFR ($0 < \Delta x_T < 1$) induced similar transition position shifts compared to equivalent FFS, the critical FFR were able to push the limits of FFS to $h/\delta_1 > 3.5$.

Such difference was not observed between BFS and BFR, which were found to be much more destabilizing than FFS and FFR. Although this result is in agreement with previously published data from incompressible flow experiments, the critical dimensions for the BFS in compressible flow were reached for lower dimensions $h/\delta_1 > 1.1$, compared to $h/\delta_1 > 1.5$ for incompressible flow. Future work should

also include further investigation on the lack of difference between equivalent BFS and BFR on the transition position.

Additionally, comparing the two data sets for FFS and BFS, a non-negligible proportion of FFS were found to have Δx_T values between 1 and 1.2, indicating possible stabilization for FFS with $h/\delta_1 < 1$. In previous studies (e.g., by Wörner et al. (2003)), FFS were also found to slightly delay the transition position, for low h/δ_1 values (less than 0.5). Although introducing FFS to stabilize the boundary layer cannot be considered as a desirable or robust solution, the present results confirm that, in practical situations where surface defects cannot be avoided, FFS are to be preferred to BFS.

In the present experimental campaign, both regular gaps and gaps with uneven sides, referred to as steps and gaps or S&G, were tested and compared to the Olive and Blanchard criteria for critical gaps. Although determining whether or not regular gaps were critical could not be ascertained due to some limitations in the model, the ΔN values based on the subcritical gaps showed relatively good agreement with the ΔN model for gaps proposed by Crouch et al. (2022).

Since S&G are asymmetric, different dimensions can be used for their characterization. When comparing to the gaps' depth h/δ_1 , the upstream side of the S&G h_1/δ_1 was considered most relevant. With this choice of parameter, most critical S&G are consistent with the conjunct Olive and Blanchard criteria. Some S&G were also found to be critical because their step component, defined as $h/\delta_1 = h_1/\delta_1 - h_2/\delta_1$.

Finally, one interesting result came from a critical S&G geometry where neither the depth nor the step components could be identified as mainly responsible for the onset of transition. Stability calculations over the S&G and the different corresponding combinations of BFS or gaps showed that the S&G induced greater amplification of the TS waves than any of the other defects. Investigating S&G, such as in Rouviere et al. (2022), should therefore be included, whenever possible, to the canonical shapes of surface defects relevant to laminar–turbulent transition.

Symmetry and two-dimensionality of the aerodynamic conditions

Figure 16 shows the pressure coefficient distributions at $M = 0.77$ for all three static pressure port sections. The zero lift angle of attack was determined once the upper and lower centerline pressure distributions overlapped, while two-dimensionality of the flow over the model was confirmed by comparing the upper centerline and upper starboard pressure distributions. Based on this representative

case, flow symmetry between the upper and lower surfaces, and flow two-dimensionality over the span of the model is validated.

Determining ΔN values

Figure 17 provides a schematic representation of the process used to determine the ΔN values by combining linear stability theory calculations with experimental transition positions. In the IR measurement box, the red or blue line corresponds to the automatically detected transition front based on the temperature gradient.

All stability calculations are based on the CFD RANS results. It is necessary to evaluate and quantify the uncertainty on the calculation of the N factor in the smooth configuration, and thus derive an uncertainty for the ΔN values obtained. An approach close to a Monte Carlo method is therefore implemented, necessarily limited by the complexity of the calculations. The RANS simulation is designed to reproduce the flow generating conditions found in the experiments. An uncertainty is then quantified by measuring the mean deviation between the pressure coefficient derived from the numerical simulation and the measured one. This value is considered the uncertain part of our simulation. A number of random pressure coefficient curves (25 random pressure distributions) are generated, to which some random noise is added (based on the mean deviation determined above) and then the curves are smoothed. A linear stability calculation is then performed for each pressure distribution. A series of N factor curves is then obtained, allowing us to assess the uncertainty in the calculation of the reference N factor curve. Figure 18 shows the results obtained for 25 pressure distributions for the $M_0 = 0.6884$ and $Pi_0 = 130$ kPa configuration. Applied to all test configurations, this uncertainty estimation procedure gives an error margin of ± 0.2 in the determination of the N factor.

These calculations also enabled to determine an uncertainty on the displacement thickness δ_1 at ± 2 μm . Based on this uncertainty, the uncertainty of the surface defects' relative height h/δ_1 was determined to be at ± 0.1 .

LNS simulation of a low- h/δ_1 FFS

Using the same procedure to obtain results shown in Fig. 15, LNS simulations (see details in Rouviere et al. (2023)) were performed on a flat plate for an FFS of $h = 50$ μm with $h/\delta_1 = 0.37$, which is in the range of the stabilizing FFS presently observed. Calculations for the corresponding smooth case were also performed, as a comparison. Results from these

simulations are shown in Fig. 19. Since the method is based on the two-dimensional incompressible Navier–Stokes equations, only the lowest experimental Mach number was simulated ($M_0 = 0.295$, $Pi_0 = 60$ kPa, $Ti_0 = 295$ K). The experimental case gives a ΔN of 0.48 which is consistent with the LNS result. In this incompressible flow, the FFS still destabilizes the boundary layer, shown by the greater maximum N factors far downstream of the FFS (i.e., $(x - x_d)/\delta_{1,d} > 100$). However, locally, the geometry of the FFS induces a significant adverse pressure gradient immediately upstream of the step, shown as the large increase in N factor for the negative streamwise coordinates, followed by a favorable pressure gradient at the top of the step, indicated by the decrease in N factor.

Several h/δ_1 for $Re_{\delta_1} = 1100$ have been simulated in order to assess the stabilizing effect of a FFS. Figure 20 exhibit a stabilization extend immediately downstream the defect. The N factor with the defect is lower than the N factor in the smooth condition. This stabilization extend is even greater as the h/δ_1 decreases. The threshold h/δ_1 around 0.2 found in Wörner et al. (2003) seems to be confirmed.

In addition, FFS with this h/δ_1 was found to be even more stabilizing in the experiments, one additional stabilizing component could be due to compressibility effects. In particular, the attenuation of the boundary layer instabilities (N factor reduction) downstream of the step could more significantly outweigh the amplification of the instabilities (N factor increase) upstream of the step, resulting in lower N factors far downstream of the step, compared to the smooth case.

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Author Contributions JM, FM, and OV wrote the main manuscript text. FM, OV, and MF designed the model. JM, MF, and FM performed the test. AR performed the CFD calculation and the LNS computations. JM performed all the data analysis. All authors reviewed the manuscript.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval Not applicable.

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